

MAKKAUR FYR, Norway

WMO: 010920

Lat: 70.707N

Lon: 30.079E

Elev: 30

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.8	12.3	0.8	5.7	10.8	5.1	7.1	13.2	56.9	8.3	53.8	12.8	23.1	220	0.887

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	69.1	57.7	64.1	55.8	59.8	53.3	58.9	67.7	56.3	63.2	53.7	59.0	12.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.3	62.7	62.7	52.0	57.7	59.4	50.3	54.0	56.6	25.7	68.0	24.0	63.5	22.4	59.2	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
43.2	36.4	31.3	4.4	76.4	4.8	5.7	0.9	80.5	-1.9	83.8	-4.7	87.1	-8.2	91.2
			3.9	61.4	4.3	3.7	0.8	64.1	-1.7	66.3	-4.1	68.4	-7.3	71.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	24.8	24.6	26.9	32.6	39.3	44.9	51.4	50.7	46.6	38.7	31.6	28.6
	DBStd	11.06	6.46	6.79	5.54	5.67	5.57	5.67	6.48	5.07	5.06	5.52	5.02	5.52
	HDD50	5036	780	710	715	521	335	176	54	51	129	351	552	662
	HDD65	10292	1245	1130	1180	971	796	602	426	446	553	814	1002	1127
	CDD50	223	0	0	0	0	4	24	98	71	25	1	0	0
	CDD65	7	0	0	0	0	0	0	6	1	0	0	0	0
	CDH74	46	0	0	0	0	0	2	41	3	0	0	0	0
	CDH80	6	0	0	0	0	0	0	6	0	0	0	0	0
Wind	WSAvg	14.0	18.0	18.6	18.1	13.9	11.4	10.6	9.7	9.4	11.3	14.1	15.9	17.7
Precipitation	PrecAvg	22.8	1.8	1.6	1.5	1.3	1.2	1.9	2.3	2.6	2.2	2.9	1.8	1.6
	PrecMax	27.2	2.5	2.5	2.3	2.1	2.4	3.6	5.5	5.7	3.6	4.5	3.1	2.2
	PrecMin	20.3	0.6	0.6	0.6	0.4	0.5	0.3	0.5	1.1	1.3	1.1	0.4	0.5
	PrecStd	2.2	0.6	0.5	0.5	0.5	0.5	1.1	1.4	1.3	0.6	0.9	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	40.4	41.4	49.0	60.0	69.5	79.3	72.5	62.7	54.2	46.3	43.2
		MCWB	39.1	37.2	36.7	42.8	48.6	57.1	62.7	60.0	54.6	49.0	42.7	39.8
	2%	DB	38.5	38.3	39.2	45.3	54.3	62.1	71.9	66.2	59.7	50.9	42.8	40.3
		MCWB	35.4	35.5	35.7	40.0	46.2	53.8	58.6	57.3	53.2	46.5	40.0	37.2
	5%	DB	36.1	36.3	36.7	42.2	50.8	57.1	66.5	61.7	56.8	48.6	40.1	37.9
		MCWB	33.2	33.6	33.6	37.9	44.5	50.0	57.3	54.7	51.4	44.8	37.5	34.9
	10%	DB	34.1	34.0	35.0	39.9	47.5	52.0	61.2	57.7	53.9	46.2	38.3	36.1
		MCWB	31.5	31.4	32.1	36.3	42.0	46.9	54.8	52.5	49.2	43.2	35.8	33.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	38.2	37.8	43.4	49.8	59.1	64.1	61.2	56.1	49.4	43.1	40.3
		MCDB	41.9	40.0	40.6	48.6	57.4	67.8	77.3	70.4	61.9	53.4	45.8	43.2
	2%	WB	35.7	35.5	35.7	40.4	47.1	54.6	60.1	57.8	53.8	46.8	40.3	37.5
		MCDB	38.1	37.7	38.8	44.7	53.6	61.8	70.0	64.8	58.8	50.3	42.8	40.3
	5%	WB	33.5	33.8	33.8	38.2	44.8	50.3	57.5	55.1	51.6	45.2	37.7	35.3
		MCDB	35.9	36.3	36.5	41.7	50.7	56.3	65.8	60.3	56.1	48.2	39.9	37.7
	10%	WB	31.7	31.7	32.2	36.6	42.6	47.2	54.4	52.9	49.7	43.5	36.1	33.6
		MCDB	33.9	33.9	34.7	39.4	47.0	51.3	60.8	57.1	53.4	46.3	38.3	35.9
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.1	7.3	6.8	6.6	6.5	8.0	7.0	6.1	5.6	6.6	7.6
		MCDBR	9.8	8.8	7.3	7.7	12.7	15.9	18.8	15.4	10.0	6.5	7.1	8.6
		MCWBR	8.6	7.9	6.6	5.4	7.8	9.6	9.3	8.5	6.3	5.1	6.5	7.6
	5% WB	MCDBR	9.5	8.8	7.1	7.1	12.3	14.9	17.6	14.2	9.1	6.0	6.9	8.5
		MCWBR	8.7	8.2	6.6	5.5	7.8	9.4	9.8	8.6	6.3	5.0	6.6	7.7
Clear-Sky Solar Irradiance	taub	N/A	0.261	0.314	0.335	0.350	0.341	0.357	0.344	0.315	0.266	N/A	N/A	
	taud	N/A	2.058	2.176	2.269	2.369	2.482	2.472	2.490	2.503	2.154	N/A	N/A	
	Ebn at Noon	0	130	210	250	262	270	259	249	222	126	0	0	
	Edh at Noon	0	12	23	30	30	28	28	24	17	10	0	0	
All-Sky Solar Radiation	RadAvg	0	67	389	897	1284	1494	1341	910	476	123	0	0	
	RadStd	0	9	32	70	147	108	120	91	45	14	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page